

User Story Specification

OpenCart Platform — Search & Category Navigation

Document ID: OC-SEAR-CAT-US | v1.0 | 18 Stories | 72 Acceptance Criteria

Module: Search

OC-SEAR-US-001

Core Search Functionality

As a site visitor, I want to search for products using keywords and filters so that I can quickly find what I'm looking for.

- AC-01** Given I am on the search page and I enter a valid keyword such as 'MacBook', When I click Search, Then matching products should be displayed.
- AC-02** Given I leave the Keywords field empty, When I click Search, Then all products are shown OR a 'no results' message appears — no crash.
- AC-03** Given no matching products exist, When I click Search, Then the message 'There is no product that matches the search criteria.' is displayed.
- AC-04** Given results are displayed, When I click a product link, Then I am redirected to the correct product detail page.
- AC-05** Given I use the top navigation search bar and click the magnifying glass, Then I am redirected to the search results page.

OC-SEAR-US-002

Search Filters & Options

As a site visitor, I want to refine my search using category filters, subcategory options, and description search.

- AC-01** Given a keyword and a selected category, When I click Search, Then only products in that category matching the keyword are returned.
- AC-02** Given 'Search in subcategories' is checked, When I click Search, Then results include products from all subcategories.
- AC-03** Given 'Search in product descriptions' is checked, When I click Search, Then products whose descriptions contain the keyword are returned.
- AC-04** Given category is 'All Categories', When I click Search, Then matching products from all categories are returned.

OC-SEAR-US-003

Search Input Boundary Values (BVA)

As a site visitor, I want the search input to correctly handle minimum and maximum character lengths.

- AC-01** Given a single character 'a', When I click Search, Then all items containing that character are returned.
- AC-02** Given exactly 2 characters 'HP', When I click Search, Then HP-related products are returned with no validation error.
- AC-03** Given exactly 255 characters, When I click Search, Then the input is accepted without truncation or error.
- AC-04** Given 256+ characters, When I click Search, Then input is truncated OR a validation error is shown — no server error.

OC-SEAR-US-004

Search Equivalence Partitioning (EP)

As a site visitor, I want the search to handle different input types (case, format, language) consistently.

- AC-01** Given uppercase 'MACBOOK', When I search, Then MacBook products are returned (case-insensitive).

AC-02 Given lowercase 'macbook', When I search, Then MacBook products are returned.

AC-03 Given mixed-case 'mAcBoOk', When I search, Then MacBook products are returned.

AC-04 Given numeric '3065', When I search, Then products matching that number in name or model are returned.

AC-05 Given alphanumeric 'LP3065', When I search, Then matching products are returned correctly.

AC-06 Given ' MacBook ' with spaces, When I search, Then whitespace is trimmed and MacBook products are returned.

OC-SEAR-US-005

Search Error Handling

As a site visitor, I want the search system to handle unexpected inputs gracefully.

AC-01 Given only whitespace in the Keywords field, When I click Search, Then the system trims to empty and returns no results.

AC-02 Given only special characters '@#\$%^&*()', When I click Search, Then no results are returned and the 'no match' message is shown.

AC-03 Given Arabic text '■■■■■■■', When I click Search, Then the input is accepted without encoding errors.

AC-04 Given a double-click on the Search button, When triggered, Then the search executes only once with no duplicate requests.

AC-05 Given a back-button press after viewing a product, When triggered, Then I return to the search results with the previous query intact.

OC-SEAR-US-006

Search UI & Visual Design

As a site visitor, I want the search page to be visually consistent and display all UI elements correctly.

AC-01 Given the search page loads, Then breadcrumb, keyword field, category dropdown, checkboxes, and Search button are all visible and aligned.

AC-02 Given the Keywords field is not focused, Then placeholder text 'Keywords' is visible in grey.

AC-03 Given the search page loads, Then the browser tab shows 'Search' and breadcrumb shows: Home > Search.

AC-04 Given the search page loads, Then the header shows logo/nav/cart and the footer shows all four columns with correct links.

AC-05 Given I view the Search button, Then its design is consistent with the overall site theme.

OC-SEAR-US-007

Search Usability

As a site visitor, I want to interact with the search form using keyboard and mouse intuitively.

AC-01 Given I click the Keywords field, Then a visible focus outline appears.

AC-02 Given I press Tab in the form, Then focus moves: Keywords → Category → Description checkbox → Search button.

AC-03 Given I click a checkbox label, Then the checkbox toggles (label is bound to input).

AC-04 Given I press Enter after typing a keyword, Then search is triggered identically to clicking the Search button.

OC-SEAR-US-008

Search Performance & Response Time

As a site visitor, I want search results to load within an acceptable time.

AC-01 Given I press F5 to refresh after a previous search, Then the same search re-executes and the same results appear.

Module: Category Navigation

OC-CAT-US-001

Category Navigation Bar

As a logged-in user, I want to see and interact with the full category navigation bar.

AC-01 Given I am logged in on the homepage, Then the navigation bar is visible with all 8 categories: Desktops, Laptops & Notebooks, Components, Tablets, Software, Phones & PDAs, Cameras, MP3 Players.

AC-02 Given I hover over any category, Then a visual style change appears and the cursor changes to a pointer.

OC-CAT-US-002

Category Page Navigation

As a logged-in user, I want to click any category and be taken to the correct page with relevant products.

AC-01 When I click 'Desktops' > 'Show All Desktops', Then the Desktops page loads with correct products.

AC-02 When I click 'Laptops & Notebooks' > 'Show All', Then the Laptops page loads with relevant products.

AC-03 When I click 'Components' > 'Show All', Then the Components page loads with products or subcategories.

AC-04 When I click 'Tablets', Then the Tablets page loads with relevant products.

AC-05 When I click 'Software', Then the Software page loads with relevant products — not an empty list.

AC-06 When I click 'Phones & PDAs', Then the page loads with relevant products.

AC-07 When I click 'Cameras', Then the Cameras page loads with relevant products.

AC-08 When I click 'MP3 Players' > 'Show All', Then the MP3 Players page loads with relevant products.

OC-CAT-US-003

Hero Banner & Featured Products

As a logged-in user, I want the homepage hero banner to function correctly and link to the right product pages.

AC-01 Given I use the banner arrows or dots, Then the slideshow responds and navigation controls work.

AC-02 Given I click a featured product image (e.g., MacBook), Then I am directed to that specific product's page.

OC-CAT-US-004

Product Actions from Category Page

As a logged-in user, I want to add products to cart, wish list, or compare list directly from a category page.

AC-01 Given I click 'Add to Cart' on a product, Then the product is added and a success message appears.

AC-02 Given I click the heart (wish list) button, Then the product is added to my wish list and the header count updates.

AC-03 Given I click the Compare icon on two or more products, Then they are added to the comparison list successfully.

OC-CAT-US-005

Category Page Advanced Features

As a logged-in user, I want to use sorting, view switching, and multi-category cart features.

AC-01 Given I select 'Price (Low > High)' from Sort By, Then products re-order from lowest to highest price.

AC-02 Given I toggle List/Grid view icons, Then the layout switches correctly and reflects immediately.

AC-03 Given I add products from Tablets and Phones & PDAs to cart, Then both appear in the Shopping Cart with correct totals.

AC-04 Given I select EUR from the currency dropdown, Then all prices update to EUR across the site.

OC-CAT-US-006

Category Breadcrumb & Navigation Flow

As a logged-in user, I want breadcrumbs to update correctly and navigation to flow logically.

AC-01 Given I click 'Tablets', Then the breadcrumb shows: Home > Tablets.

AC-02 Given I click 'Components' then 'Monitors', Then the breadcrumb shows: Home > Components > Monitors.

AC-03 Given I click the logo or Home breadcrumb, Then I return to the homepage with my login session preserved.

OC-CAT-US-007

Category Page UI & Visual Design

As a logged-in user, I want category pages to have a consistent and correct visual layout.

AC-01 Given I navigate through all 8 categories, Then all pages share the same layout: breadcrumb, sort bar, product grid, pagination.

AC-02 Given I hover over a category with subcategories, Then the dropdown is visible and navigable.

AC-03 Given I open any category page, Then the correct category name and products are displayed per the database.

OC-CAT-US-008

Category Navigation Usability

As a logged-in user, I want to navigate categories using keyboard and browser controls.

AC-01 Given keyboard-only navigation, When I Tab to 'Tablets' and press Enter, Then the Tablets page loads.

AC-02 Given I press the Back button after viewing a product, Then I return to the previous category page.

AC-03 Given I open a category link in a new tab, Then the correct page loads with my login session intact.

OC-CAT-US-009

Category Error Handling

As a logged-in user, I want the system to handle invalid navigation actions gracefully.

AC-01 Given I enter an invalid category URL, Then a friendly 'Category not found!' error page is shown and my session stays active.

AC-02 Given I double-click a category link, Then the page loads only once with no duplicate requests or freeze.

AC-03 Given I press F5 on a category page, Then the same category re-renders correctly with the same products.

OC-CAT-US-010

Currency Selector

As a logged-in user, I want to change the display currency so I can view prices in my preferred currency.

AC-01 Given I select EUR from the currency dropdown, Then all product prices update to EUR across all category pages.

Definition of Done

- ✓ All Acceptance Criteria passed
- ✓ No Critical / High defects open
- ✓ Cross-browser compatibility verified (Chrome, Edge)
- ✓ Search BVA and EP scenarios validated
- ✓ Category navigation tested for all 8 categories
- ✓ Security checks passed (SQL Injection, XSS)
- ✓ Usability and keyboard accessibility verified